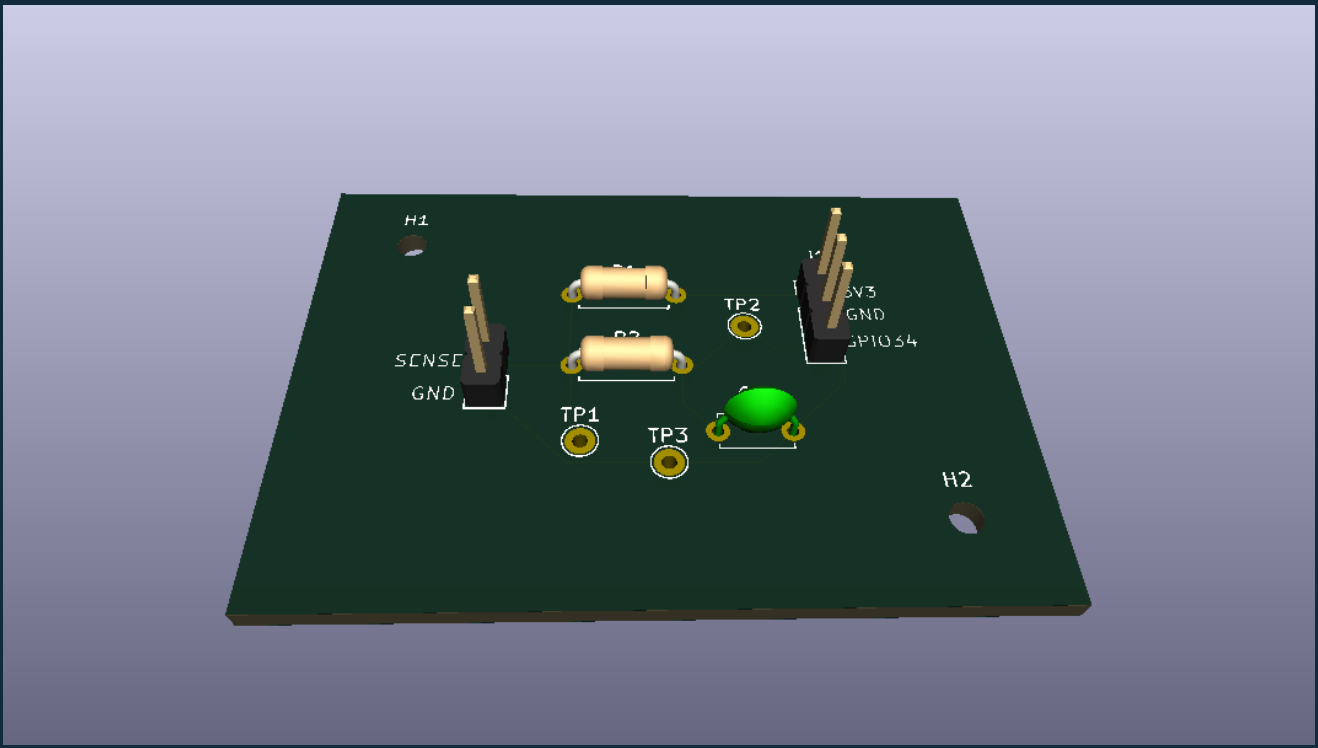
ELECTRICAL ENGINEERING / VERIFIED SOURCE EVIDENCE

THERMOBENCH

Design, PCB CAD, Embedded Firmware and Virtual Validation

Real ESP32 breadboard datasets | KiCad schematic, PCB and 3D model | LTspice circuit simulation | Wokwi firmware and hysteresis testing



ENGINEERING DESIGN & VALIDATION REPORT

Revision 2.0 | 21 October 2026 | 23 pages

Prepared for technical interviews and reproducible project handoff

Unfabricated PCB; hardware and virtual results distinctly labeled.

01 / READOUT

Project Summary

This report combines the submitted CAD source, software project and original ESP32 hardware captures.

STABLE RUN	ACQUISITION	PCB DESIGN
0.041°C	**5.00 Hz**	**50 × 35**
Sample SD, n = 641	200 ms per sample	millimeters, CAD only

ENGINEERING QUESTION

Can a simple 10 kΩ NTC voltage-divider interface be read consistently by an ESP32 ADC and documented as a repeatable sensing prototype, with a traceable PCB design?

WHAT WAS ACTUALLY DEMONSTRATED

- Breadboard ESP32 acquired four CSV recordings with **all device timestamps increasing in 200 ms steps**.
- A stationary hardware capture (641 samples over 128 s) showed **0.0412°C sample standard deviation** in firmware-equivalent temperature.
- A separate thermal recording showed a rise from an approximately **22.51°C initial 20 s average** to **30.829°C peak**, followed by cooling.
- The PCB exists as editable KiCad schematic and board files. Screenshots show **zero active DRC violations, zero unrouted pads and zero parity warnings**.

IMPORTANT VERIFICATION BOUNDARY

The breadboard assembly has real hardware evidence. The PCB was not fabricated, soldered or electrically tested. Model-derived temperature is not reference-calibrated temperature. Component-fit checks and absolute temperature-accuracy verification are outstanding.

02 / ARCHITECTURE

System Architecture

Low-voltage sensor front end → ESP32 ADC → serial measurement stream → Python dashboard and CSV analysis.



MEASUREMENT CHAIN

- R1 = 9.82 kΩ pulls the SENSE node toward the measured 3.2 V rail; the external NTC connects SENSE to GND.
- R2 = 990 Ω connects SENSE to the ADC input; C1 = 100 nF filters the ADC node to GND.
- The ESP32 outputs **ms,mv,temp_c,status** at a configured 200 ms interval; Python stores readings with a HARDWARE/SYNTHETIC source field.
- The board is an **adapter**: J1 exposes 3V3, GND and GPIO34, while J2 exposes SENSE and GND; the ESP32 itself is not mounted on this PCB.

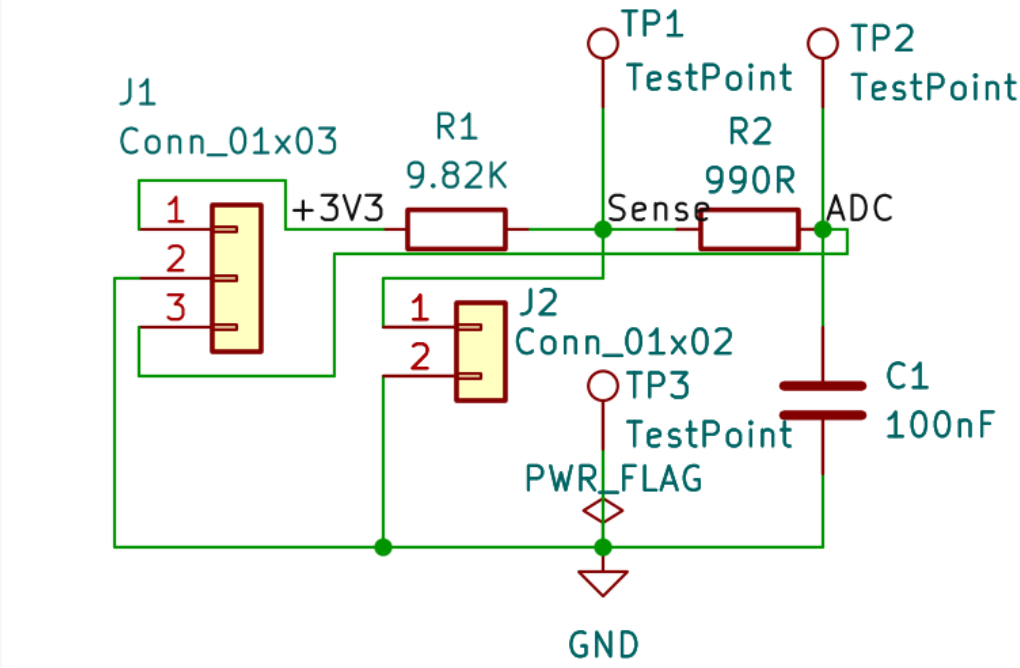
BOUNDARY BETWEEN DEMONSTRATION AND CALIBRATION

ADC readings show plausible electrical behavior, but conversion accuracy depends on ADC calibration, supply voltage, reference resistance, thermistor beta value, self-heating and a trusted reference thermometer. Only the response and repeatability were evaluated here.

03 / CIRCUIT

KiCad Schematic & Electrical Model

Actual circuit drawing cropped from the KiCad schematic PDF included in the project archive.



VOLTAGE DIVIDER

$$V_{SENSE} = V_{SUPPLY} \times R_{NTC} / (R1 + R_{NTC}) \mid R_{NTC} = R1 \times V_{SENSE} / (V_{SUPPLY} - V_{SENSE})$$

SCHEMATIC COMPONENT SCHEDULE

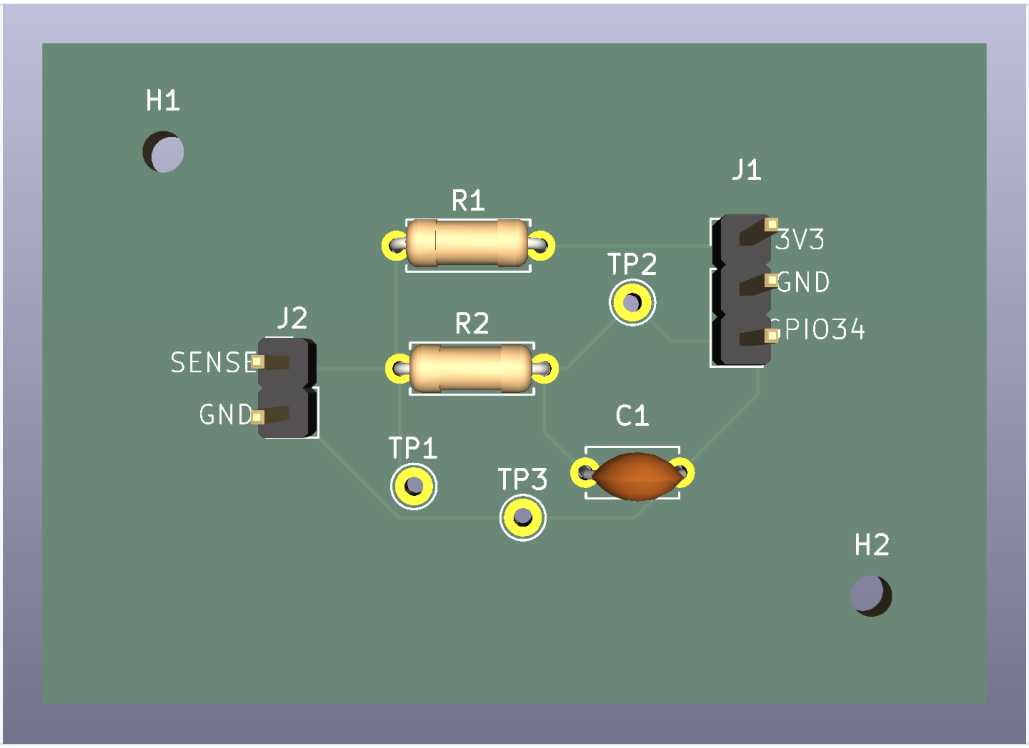
Reference	Schematic value	Purpose
R1	9.82 kΩ	Top divider resistor
R2	990 Ω	ADC input series resistor
C1	100 nF	ADC low-pass / noise filter
J1	3-pin, 2.54 mm	To ESP32: 3V3 / GND / GPIO34
J2	2-pin, 2.54 mm	External NTC: SENSE / GND
TP1, TP2, TP3	Test points	SENSE / ADC / GND

The external thermistor is represented by J2 in the PCB-adapter schematic rather than soldered directly onto the board.

04 / CAD

PCB Layout & Routing

The PCB-native file contains one Edge.Cuts rectangle, 10 footprints (including H1/H2), 22 track segments and no vias.



BOARD WIDTH

50.0 mm

From Edge.Cuts

BOARD HEIGHT

35.0 mm

From Edge.Cuts

VIAS IN FILE

0

All through-hole parts

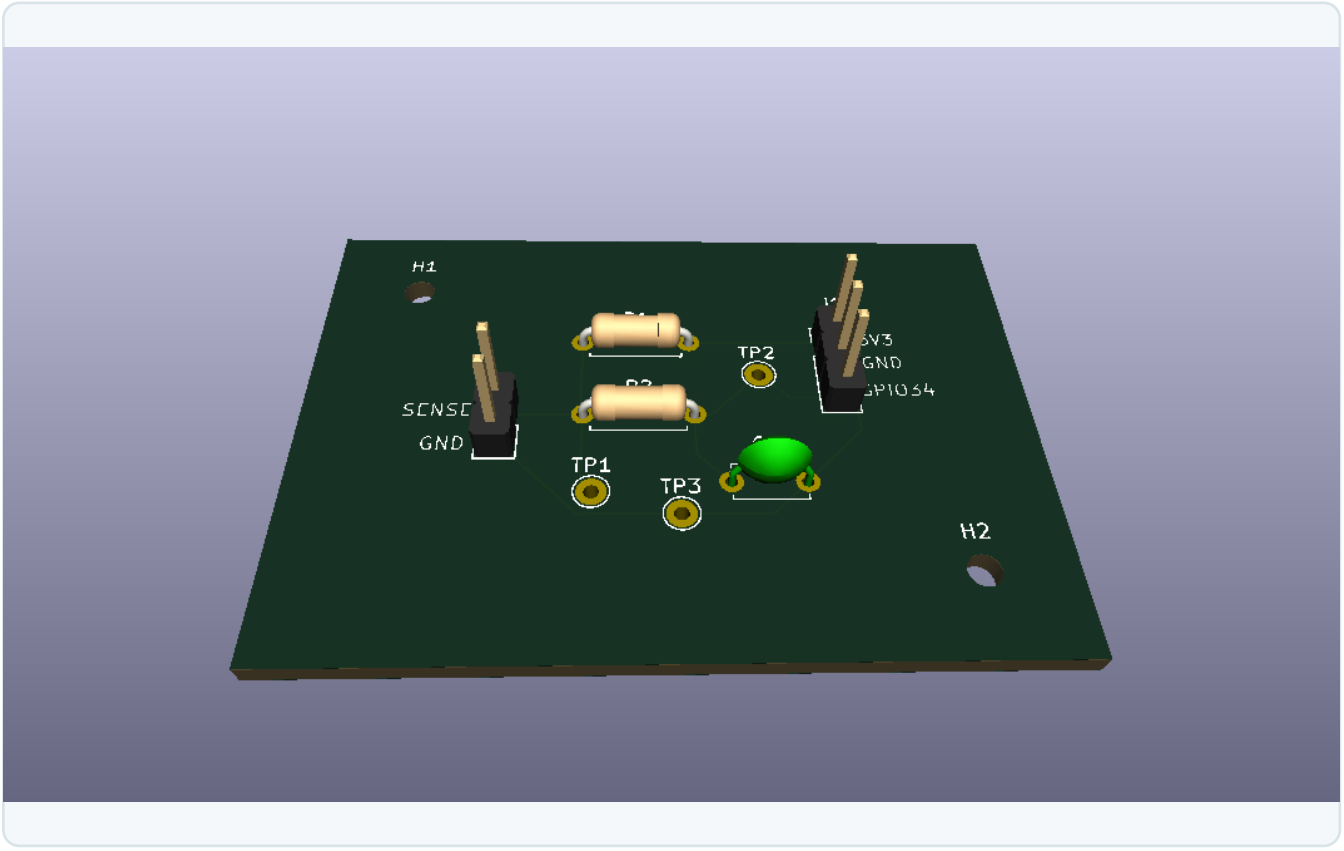
COPPER CONNECTIVITY

Net	Pads / circuit nodes	Design treatment
+3V3	J1.1 → R1.2	Supply input
SENSE	R1.1, R2.1, J2.1, TP1	Divider node
ADC	R2.2, C1.1, J1.3, TP2	Filtered ADC measurement
GND	J1.2, J2.2, C1.2, TP3	Common return

05 / VISUALIZATION

3D PCB Assembly

The KiCad 3D model is an illustrated design review—not proof that purchased components fit physically.



WHAT THE MODEL SHOWS

- Two standard axial through-hole resistors and one radial ceramic-disc capacitor.
- Two pin-header connector footprints, three test pads and two 2.2 mm non-plated M2 mounting holes.
- Silkscreen labels for 3V3, GND, GPIO34, SENSE and the test points.

PRE-FABRICATION MECHANICAL CHECKS

Before manufacturing: measure actual resistor bodies and lead spans; measure C1 body and lead pitch; confirm J1/J2 mating direction, pitch and height; verify screw-head diameter/driver clearance at H1/H2; check solder-side space and enclosure constraints. The current PCB was intentionally not ordered or assembled.

06 / FIRMWARE & SOFTWARE

Acquisition Pipeline

The supplied source repository includes ESP32 firmware, a Python dashboard, serial parsing, validation scripts and unit tests.

Layer	Implementation
Firmware: thermobench.ino	Arduino ESP32 ADC1 GPIO34; averages 16 analogReadMilliVolts readings
Serial stream	115200 baud; ms, mv, temp_c, status
Python dashboard	Live USB data visualization, source label and CSV recording
Analysis / testing	station.py, test_station.py, test_bench.py and analysis code

EMBEDDED DECISION LOGIC

```
if mv <= 100 mV:       LOW_FAULT
else if mv >= 2900 mV: HIGH_FAULT
else:                  convert V → R → model T
   model T outside 15–55°C: OUT_OF_RANGE
   model T >= 45°C:      HOT
   model T <= 43°C:      NORMAL
```

SOURCE-TO-HARDWARE CAVEAT

The archived **thermobench.ino** still has nominal constants **VS_MV = 3200** and **RF = 10000**. Earlier bench work used measured values near 3190–3200 mV and R1 ≈ 9820 Ω, but the uploaded ZIP does not establish the exact binary flashed for the recorded trials. Preserve this version distinction in interviews; do not present the archived source as a byte-for-byte capture of the flashed firmware.

07 / EVIDENCE

Hardware Dataset Inventory

Four CSV captures were found under project/captures. All contain source=HARDWARE and consistent 200 ms device-time intervals.

Capture ID	Samples	Span	Observed pattern
22:47:27	16	3.0 s	Short cooling observation
22:47:52	133	26.4 s	Initial warming response
23:47:15	641	128.0 s	Stationary repeatability
23:52:31	544	108.6 s	Warm-up followed by cooling

WHAT THESE DATASETS DO AND DO NOT PROVE

- All four captures maintain 200 ms increments between adjacent device timestamps; no gaps appeared in the uploaded CSVs.
- Each reported **NORMAL** status for its entire recording. No actual fault-injection capture is included.
- The **128.0 s** stationary trial is used for repeatability statistics; warming trials are not valid stationary-noise datasets.
- Measured temperature is a **model-equivalent estimate** based on NTC nominal values and ADC voltage; a calibrated reference trace is not included.

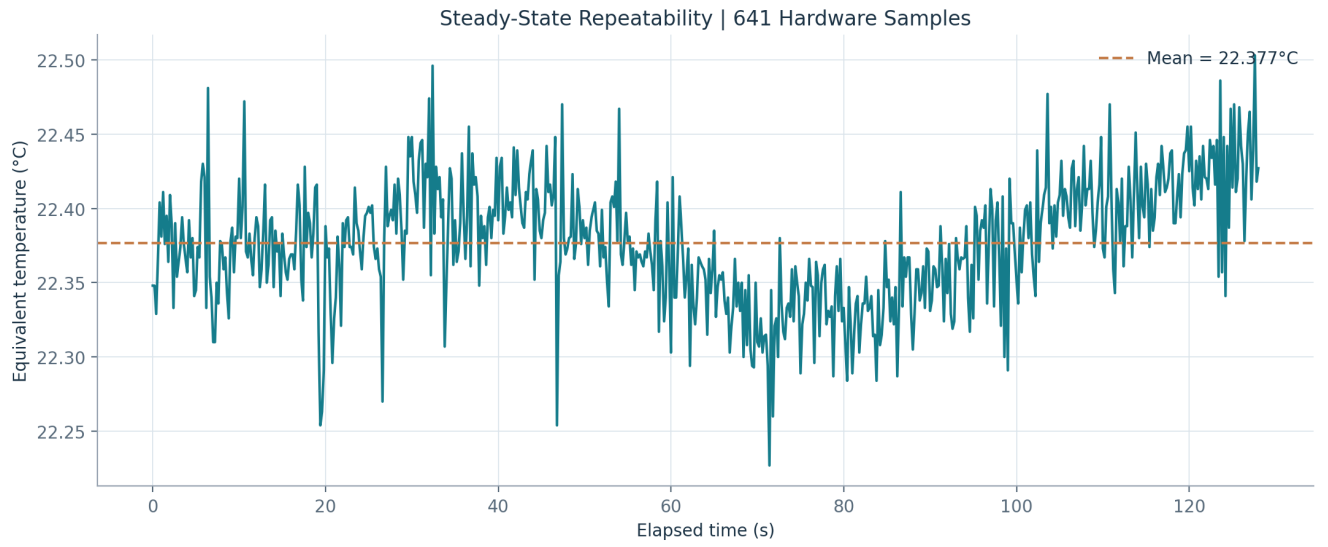
HANDLING SYNTHETIC RECORDS

The original software archive also contains demo CSVs, SPC teaching plots and synthetic sweep files. Those are retained only as labeled software examples in the consolidated ZIP, and no synthetic observations are included in the real-hardware statistics in this report.

08 / REPEATABILITY

Steady-State Validation

Real ESP32 run: hardware_20261019_234715_916346.csv. Statistics recomputed from all 641 rows.



MEAN	SAMPLE SD	VOLTAGE SD
22.377°C	**0.0412°C**	**1.480 mV**
Equivalent temperature	n = 641	n = 641

SCREENING RESULT

The project-defined screening criterion was **sample SD ≤ 0.15°C** for an untouched, stable input. The observed sample SD is **0.0412°C**, approximately 27% of that limit. This is a pass for the stated **repeatability screening**, not for thermal accuracy.

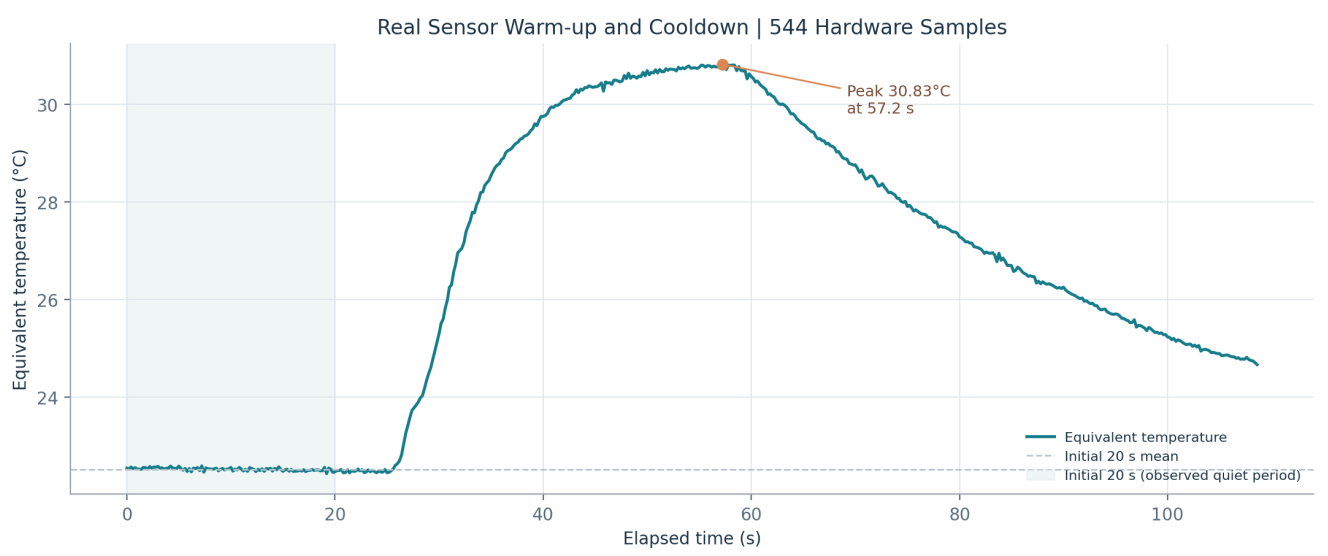
ADDITIONAL OBSERVATIONS

Metric	Observed
Recorded duration	128.0 seconds
Temperature span	22.227 to 22.503°C
Mean reported voltage	1703.112 mV
Device timestamp increments	640 / 640 at exactly 200 ms

09 / DYNAMIC RESPONSE

Thermal Warming & Cooling

Real ESP32 run: hardware_20261019_235231_403927.csv. No smoothing or synthetic samples.



INITIAL 20 S MEAN	PEAK	PEAK RISE
22.513°C	30.829°C	+8.316°C
Model-estimated	At 57.2 s	Above initial mean

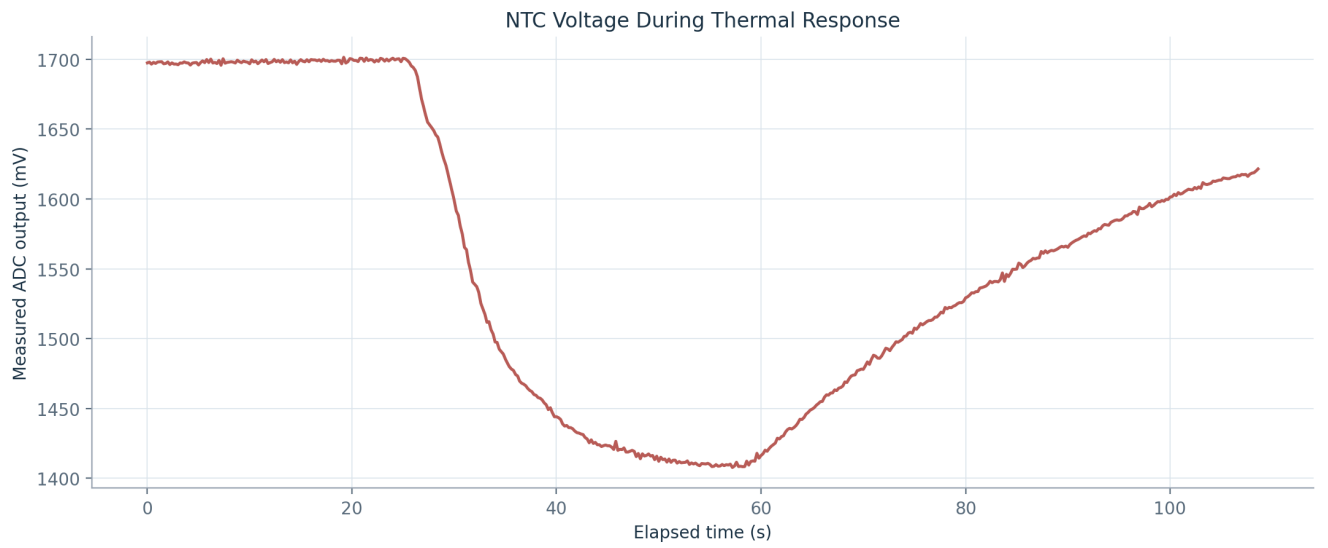
INTERPRETATION

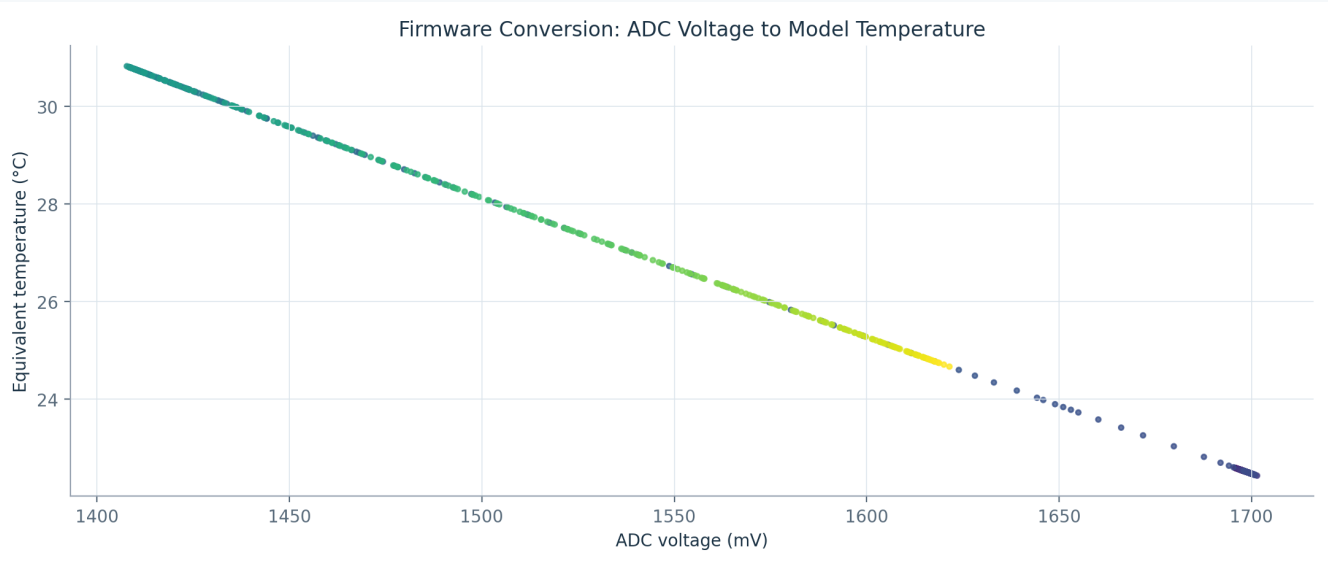
- Voltage decreased while the thermistor warmed, as expected for an NTC in the lower leg of the divider.
- A cooling phase followed the peak; at the end of the 108.6 s recording, reported temperature was **24.661°C**.
- The final reading remained about **2.15°C** above the initial-20-second mean, so full return to baseline was **not demonstrated**.

10 / ELECTRICAL RESPONSE

Voltage-Temperature Relationship

Real hardware voltage recorded during the same warm-up and cooldown run; no independent thermometer was attached.





The inverse relationship follows the divider topology and the firmware conversion function. The scatter plot is **not independent evidence of temperature accuracy**: temperature is calculated from the voltage on its horizontal axis.

11 / QUALITY GATES

Verification Register

Verification is scoped to what the uploaded files and the observed KiCad screens actually establish.

Test / claim	Evidence	Assessment
KiCad schematic connectivity	Source .kicad_sch, earlier ERC UI	Screened; ERC reported pass
PCB edge & routing	PCB source: one 50×35 mm outline; 22 segments	Verified by parsing source
PCB design-rule screening	KiCad DRC UI: 0 active violations	Passed configured checks
PCB unconnected and parity	KiCad UI: 0 / 0	Passed configured checks
Device sampling	Four raw HARDWARE CSVs	All increments 200 ms
Stationary repeatability	641-sample run, SD 0.0412°C	Pass vs 0.15°C screening
Sensor response	544-sample warming/cooling CSV	Response demonstrated
ADC absolute voltage accuracy	DMM around 1.58 V vs ADC around 1.609 V	Borderline; no calibration proof
Thermal absolute accuracy	No traceable thermometer included	NOT verified
Hardware fault and HOT states	No fault-injection hardware CSV included	NOT verified
Fabricated PCB operation	No built/assembled PCB included	NOT verified

Five DRC rule categories are configured as ignored in the saved project screenshot. Zero active violations does not guarantee complete mechanical fit, completeness of every skipped check, or manufacturability.

12 / HANDOFF

Interview & Project Handoff

What to show, what to say and where to find the underlying design evidence in the accompanying ZIP.

ONE-MINUTE PROJECT EXPLANATION

"I built an ESP32-based thermistor validation prototype, starting from a voltage divider and low-pass ADC input. I measured the breadboard supply and test nodes with a multimeter, then used GPIO34 firmware to stream real voltage and model-based temperature data into a Python dashboard. I recorded a 128-second steady-state run with 0.041°C sample standard deviation and a separate warming/cooling response. I then created the KiCad schematic, routed a 50 × 35 mm adapter PCB with test points and mounting holes, and checked its design rules. The breadboard was tested; the PCB remains an unfabricated CAD design, and absolute temperature accuracy is still unverified."

PRIMARY PORTFOLIO FILES

Folder	Contents / intended use
01_Report	This PDF; summary of design and measured results
02_KiCad_Source	Editable .kicad_pro, .kicad_sch, .kicad_pcb
03_CAD_Visuals	3D top / angled views; schematic, layer PDFs
04_Firmware_and_Python	ESP32 sketch, dashboard, analyzer, source tests
05_Hardware_CSV	Four original HARDWARE captures
06_Analysis	Fresh graphs, JSON metrics, concise methodology
07_Reference_Exports	Optional Gerbers; historical design & synthetic examples

NEXT CREDIBLE VALIDATION MILESTONE

A reference thermometer with known uncertainty and a controlled comparison setup would make an absolute temperature-accuracy claim possible. Only after mechanical part checks and physical PCB assembly should the PCB itself be called hardware-validated.

12 / LTSPICE

Circuit Simulation & Correlation

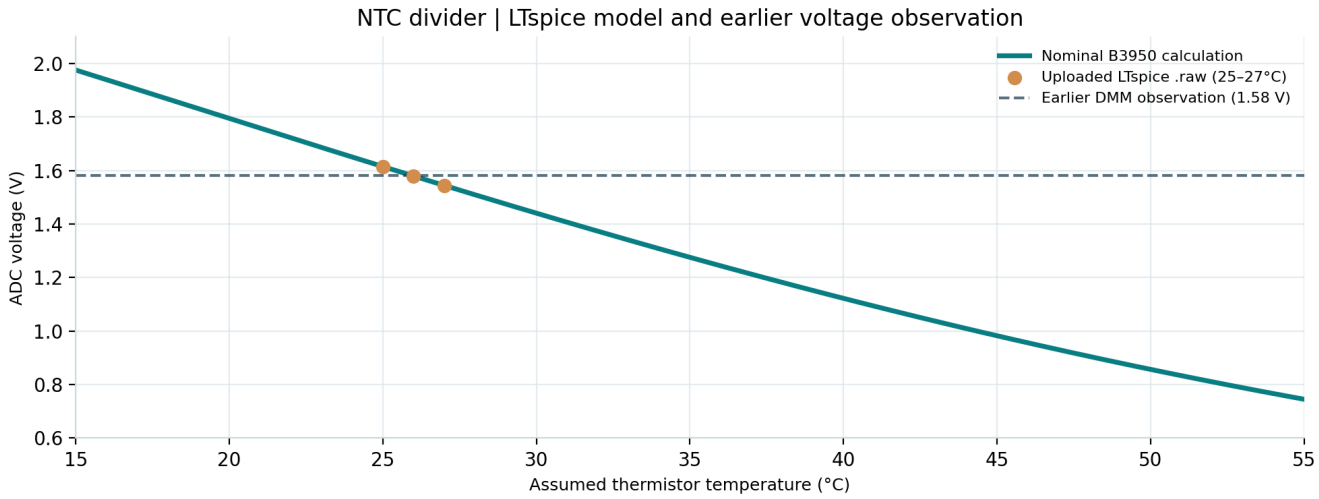
User-run .cir, .log and .raw files included with this project

LTSPICE TEMP STEPS	AC SWEEP POINTS	CUTOFF (MODEL)
3	501	267.7 Hz
25°C, 26°C, 27°C	1 Hz through 100 kHz	25°C nominal NTC model

01 / MODEL AND SIMULATION SCOPE

The uploaded LTspice source models a 3.2 V supply, R1 = 9.82 kΩ, a nominal 10 kΩ B3950 NTC sensor, R2 = 990 Ω and C1 = 100 nF. GPIO34 is represented as a high-impedance analog input, not a complete ESP32 ADC electrical model.

02 / TEMPERATURE-DEPENDENT VOLTAGE DIVIDER



Model-derived temperatures are assumptions in the simulation. An independent reference thermometer was not recorded with the earlier 1.58 V breadboard observation; no temperature calibration is claimed.

03 / SOURCE EVIDENCE

LTspice .cir files preserve the model and operating point; original .raw files preserve waveform/frequency data; original .log files preserve solver and step results. Graph datasets were extracted directly from the uploaded .raw files.

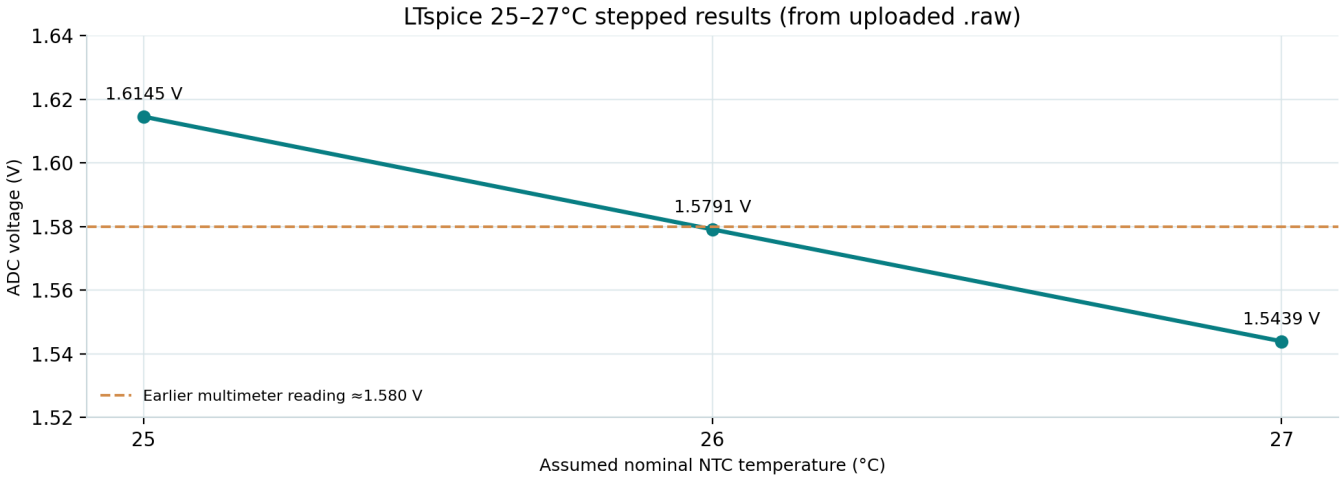
13 / TEMPERATURE

Voltage Predictions vs Breadboard

Real multimeter observation compared with nominal LTspice steps

01 / VERIFIED LTSPICE STEP VALUES

ASSUMED NTC TEMP	LTSPICE ADC	ORIGIN
25°C	1.614531 V	Actual .raw / .log
26°C	1.579103 V	Actual .raw / .log
27°C	1.543931 V	Actual .raw / .log



02 / MEASURED-VERSUS-MODELED OBSERVATION

An earlier breadboard measurement on the actual thermistor was about 1.580 V at the ADC input (3.20 V supply). A nominal 26°C model produces 1.579103 V, a difference of about 0.90 mV. This means the observed voltage is consistent with the model if the thermistor was near 26°C; it does not establish its actual temperature.

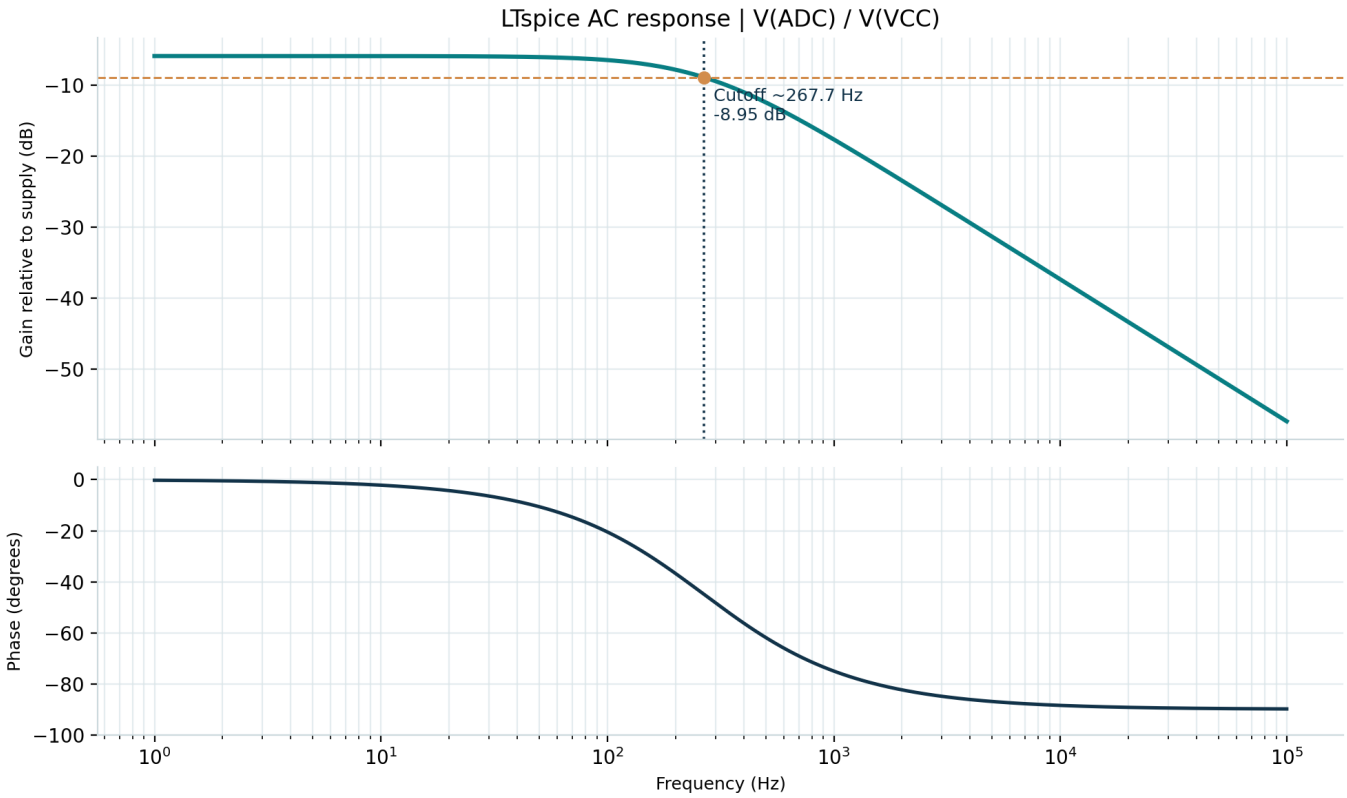
Independent temperature calibration remains pending. The uploaded LTspice netlist has a stale comment about five steps, but the saved directive and raw file actually contain only 25, 26 and 27°C.

14 / FREQUENCY RESPONSE

ADC Input Low-Pass Filter

LTspice AC analysis of R2 and C1, including divider source impedance

01 / ACTUAL SIMULATED RESPONSE



DC TRANSFER GAIN	EQUIVALENT R	-3 DB FREQUENCY
-5.94 dB	**5945 Ω**	**267.7 Hz**
9.82 kΩ / 10 kΩ divider	R2 + (R1 || RNTC)	Nominal circuit at 25°C

The correct LTspice trace is V(ADC)/V(VCC) (do not use a nested dB() call in the AC waveform viewer). Its high-frequency attenuation and phase shift are consistent with a first-order low-pass filter. Simulation does not establish real EMI performance or ADC noise reduction.

15 / HANDOFF

Simulation Evidence & Interview Notes

File map, limitations and a short explanation for technical interviews

01 / HOW TO REPRODUCE

TEMPERATURE

Open ThermoBench_LTspice.cir, run and plot V(adc). Open the SPICE Error Log for the three .meas voltages.

FREQUENCY

Open ThermoBench_Filter_AC.cir, run and add V(ADC)/V(VCC) to the AC waveform viewer.

SOURCE DATA

Read the uploaded .raw/.log files, or examine the 3-step and 501-point CSV exports supplied in the ZIP.

DOCUMENT

Use the LTspice PNG graphs along with the KiCad schematic, the PCB 3D images and hardware CSV plots.

02 / INTERVIEW-READY EXPLANATION

"I modeled my NTC voltage-divider and ADC filter in LTspice. I ran temperature-dependent simulations at 25, 26 and 27 degrees Celsius, and checked the predicted ADC voltages against an earlier multimeter observation. I also simulated the RC frequency response; its nominal cutoff was about 268 hertz. The circuit behavior is consistent with the model, but I still need an independent temperature reference to establish accuracy."

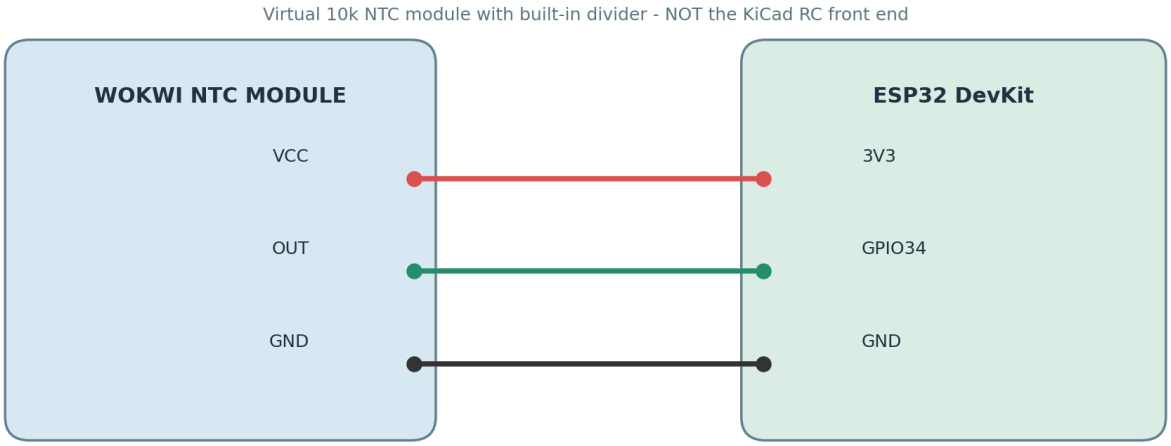
03 / REMAINING LIMITATIONS

PCB not fabricated or bench-verified. Reference thermometry and calibrated ADC comparison remain pending. The existing report records five ignored KiCad DRC check categories. Inferred temperature/model fit must not be stated as measured temperature accuracy.

Wokwi - Verified ESP32 Firmware Bench

User-executed Wokwi runs; virtual sensor module directly connected to the ESP32 ADC input.

01 / VIRTUAL WIRING



Wokwi NTC module	ESP32 DevKit	Purpose
VCC	3V3	3.3 V virtual supply
GND	GND.1	Ground reference
OUT	D34 / GPIO34	ADC1 analog input

02 / WHAT THIS SIMULATION PROVES

The supplied sketch.ino boots in Wokwi and reports ms,mv,temp_c,status at nominal 200 ms intervals. The thermistor temperature slider drove measured changes in the ADC voltage and calculated temperature. The experiment exercises virtual sensor acquisition, Beta-model conversion and state transitions.

Evidence boundary. Wokwi uses an internal 10 kOhm divider and does not reproduce the KiCad analog circuit with R1 = 9.82 kOhm, R2 = 990 Ohm and C1 = 100 nF. LTspice covers that electrical model. All Wokwi values are virtual, not physical calibration or fabricated-PCB results.

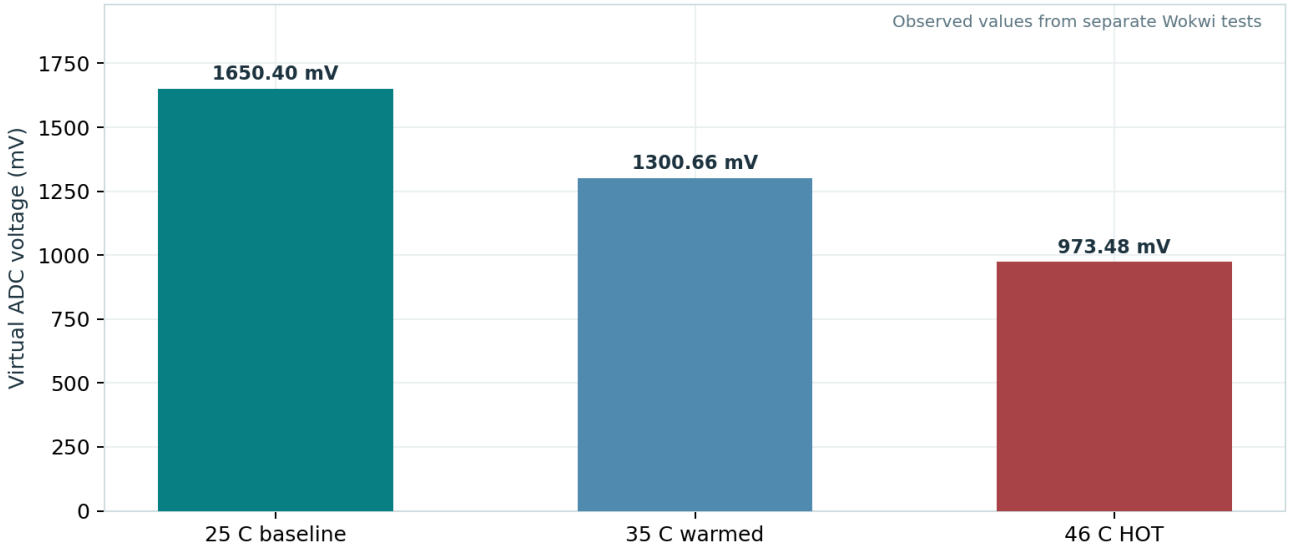
25, 35 and 46 C - observed response

Recorded values from user-shared Wokwi serial monitor; distinct test sessions are not spliced into a continuous time series.

01 / SELECTED ACTUAL VIRTUAL OBSERVATIONS

Temperature setpoint	Serial-reported value	Status
25 C baseline	1650.40 mV / 24.989 C	NORMAL
35 C stabilized	1300.66 mV / 35.000 C	NORMAL
46 C hold	973.48 mV / 45.987 C	HOT

02 / LOWER VOLTAGE AT HIGHER TEMPERATURE



03 / CONTROL AND REPORTING VALIDATION

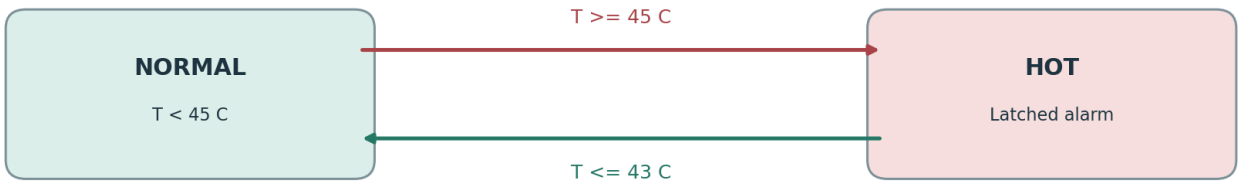
At 25 C the sketch reported 24.989 C / NORMAL. In the 35 C run, it changed from 24.989 C to 34.902 C and then 35.000 C. In the 46 C run, the alarm switched to HOT at 45.386 C and remained HOT during the 45.987 C hold. Timestamps incremented by 200 ms across logged settled intervals. These values are observed simulator outputs, not predicted values.

Source: `Wokwi_Observed_Log_Excerpts.csv` and original screenshots. CSV contains selected exact logged rows, not the entire serial session.

HOT Alarm and Hysteresis - observed

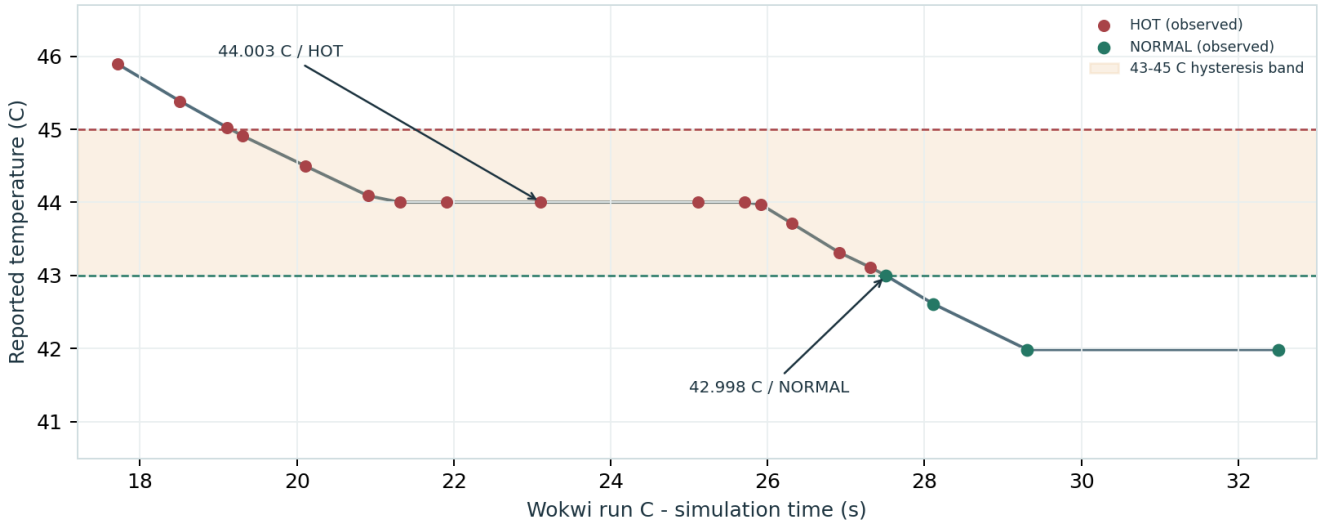
The 2 C hysteresis band was explicitly exercised in Wokwi on the cooling path.

01 / FIRMWARE STATE LOGIC



OUT_OF_RANGE and voltage faults are distinct status outputs; current firmware clears the HOT latch on faults.

02 / 44 C HOLD AND 43 C RESET

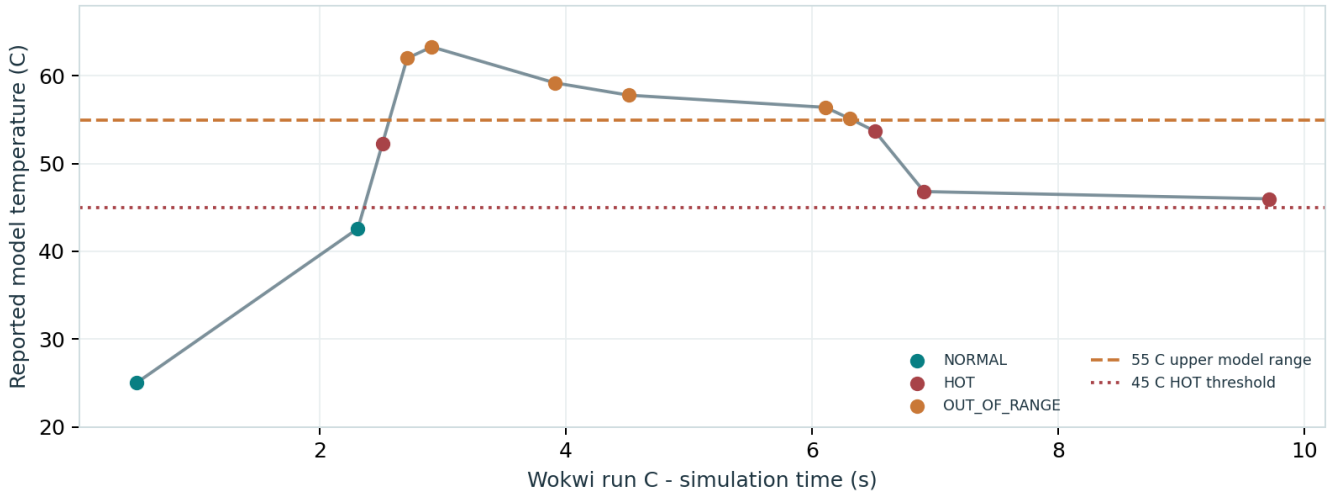


Observed temp	Status	Interpretation
45.386 C	HOT	Over 45 C turns the alarm on
44.003 C	HOT	Within hysteresis band - stays on
43.112 C	HOT	Just above reset - stays on
42.998 C	NORMAL	At or below 43 C - resets

Out-of-Range Event - captured, not hidden

A rapid setpoint change caused simulated temperature excursions above the configured 55 C software range.

01 / OBSERVED OVER-RANGE BEHAVIOR



Time (ms)	Reported temperature	Reported state
2510	52.301 C	HOT
2710	62.007 C	OUT_OF_RANGE
2910	63.304 C	OUT_OF_RANGE
6510	53.696 C	HOT

02 / CORRECT INTERPRETATION

The OUT_OF_RANGE branch fired as coded. However, the current Wokwi sketch clears the HOT latch when an over-range or voltage-fault condition occurs. A production alarm should specify how faults affect operator warnings and whether an alarm must remain latched after sensor recovery. This is a documented improvement opportunity, not a fully verified safety feature.

No separate low-voltage or high-voltage fault-injection test has been evidenced; only OUT_OF_RANGE is observed.

Cross-Tool Verification Summary

What is supported by actual source, what is only simulated, and what remains unverified.

01 / TOOL-TO-EVIDENCE MAP

Evidence source	Supported claim	Boundary
Physical ESP32 + DMM	Voltage/thermal response; 641-sample steady-state SD = 0.0412 C	Breadboard only; no reference temperature accuracy
KiCad 10 native CAD	Editable schematic; routed 50 x 35 mm PCB; 3D model; reported 0 active DRC errors	Unfabricated; five ignored DRC categories and fit checks
LTspice user-run files	1.5791 V simulated at 26 C; 267.7 Hz nominal model filter cutoff	Nominal NTC model, not calibration
Wokwi user-run logs	25/35/46 C response, HOT activation, 44 C hold, below-43 C reset	Virtual NTC internal divider, not analog PCB
Python / firmware files	Serial parsing, dashboard, analysis code, tests	Not all hardware firmware binaries traceable

02 / ENGINEERING CONCLUSION

The project now demonstrates a traceable engineering chain: analytical modeling -> circuit simulation -> embedded virtual testing -> physical breadboard measurements -> PCB CAD documentation. These are distinct forms of evidence. The hardware temperature-accuracy claim and operation of a manufactured PCB are deliberately excluded.

Reproduction, Evidence Files and Handoff

Source files, key tests, open limitations, and a short interview explanation.

01 / EXACT SOURCE FILES

Folder	Contents
02_KiCad_Source	Editable .kicad_pro, .kicad_sch and .kicad_pcb
03_CAD_Visuals	Schematic PDF, PCB copper/outline plots and 3D images
05_Hardware_CSV / 06_Analysis	Four measured ESP32 captures + recomputed graphs
09_LTspice_Simulation	Original .cir, .raw, .log + Bode and sweep graphs
10_Wokwi_Simulation	Verified sketch.ino, diagram.json, selected logs, figures, screenshots

02 / REPEAT THE WOKWI TEST

Open a new ESP32 Wokwi project. Paste the supplied sketch.ino and diagram.json; start the simulator. Confirm 25 C / NORMAL; change to 35 C / NORMAL; move to 46 C / HOT. Cool to 44 C and verify the alarm remains HOT; cool below 43 C and verify NORMAL. Finally return to 25 C. Store any future full serial export alongside the selected rows included here.

03 / SHORT INTERVIEW EXPLANATION

"I designed and tested an ESP32 NTC-sensing prototype. LTspice let me predict divider and RC-filter behavior, while Wokwi let me exercise the virtual firmware at 25, 35 and 46 degrees. I verified the HOT latch remained active at 44 degrees and reset near 43 degrees during cooling. I also documented a simulated out-of-range event. The real breadboard provided the separate measurement dataset; the custom KiCad PCB remains an unfabricated design."

04 / NEXT REAL VALIDATION

Reference thermometer, ADC calibration, actual PCB fabrication/assembly, independent hardware fault tests, and mechanical screw/head-clearance checks.